From Scalar Obstruction to Geometric Classification

A Geometry Audit for Candidate Finite-Time Singularities in Three-Dimensional Navier–Stokes

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Abstract

This note develops the next stage of a state-first investigation of finite-time singularity in the three-dimensional incompressible Navier–Stokes equations. Earlier work showed that the natural singularity observable $Q(t) = \|\boldsymbol{\omega}(t)\|_\infty$ cannot support a universal positive autonomous scalar feedback law, and that the first natural strain-aware enrichment exposes pressure, nonlocality, concentration and higher-order information rather than closing. The present note asks what can be done after that information audit: can candidate geometric configurations be classified into amplifying, neutral, and depleted families, and can broad families be ruled out without guessing a particular singular geometry? We formulate a geometry-state language based on vorticity direction, strain eigenstructure, alignment, concentration scale, directional coherence, and pressure/nonlocal coupling. We derive exact local identities and necessary scaling balances, define a hierarchy of geometric exclusion tests, and formulate a candidate blow-up class as a set of configurations capable of sustaining positive stretching, concentration, and persistence against viscous depletion. The note does not claim a solution of the Millennium problem, nor does it prove that every successful proof must identify a singular geometry. Its main contribution is a rigorous research framework: replace configuration guessing by systematic geometric elimination, with the eventual goal of narrowing any possible finite-time singularity to a sharply constrained normal form.

1 Status and scope

This note is a direct continuation of the scalar-closure investigation. The preceding version, *Vorticity Magnitude Is Not a Closed Singularity State: A Structural Obstruction and an Information Audit for Three-Dimensional Navier–Stokes*, established two methodological facts. First, a universal positive autonomous feedback depending only on $Q = \|\boldsymbol{\omega}\|_\infty$ is incompatible with broad classes of smooth dynamics, including small-data decaying solutions. Second, the natural strain-aware global production variable

$$P(t) = \int_{\mathbb{T}^3} \boldsymbol{\omega}^T S \boldsymbol{\omega}\, dx$$

does not close: differentiating P produces pressure-Hessian, rotational, diffusive, and higher-order terms. A concentration scale $L^2 = E/D$ likewise does not close.

The present note therefore changes the question. Rather than asking whether a scalar state is sufficient, we ask whether the geometric information forced by the equations can be organized into *families of configurations*, and whether those families can be progressively eliminated or retained as candidates for singularity.

The claim boundary is deliberately conservative. We do not prove global regularity, finite-time blow-up, or a universal geometric normal form. We establish exact identities, necessary-condition logic, and a research program for geometric classification.

2 Navier–Stokes and the local amplification mechanism

For incompressible flow,

$$\partial_t u + (u\cdot\nabla)u = -\nabla p + \nu\Delta u + f, \qquad \nabla\cdot u = 0. \tag{1}$$

The vorticity equation is

$$\partial_t \boldsymbol{\omega} + (u\cdot\nabla)\boldsymbol{\omega} = (\boldsymbol{\omega}\cdot\nabla)u + \nu\Delta\boldsymbol{\omega} + \nabla\times f. \tag{2}$$

Write

$$\nabla u = S + \Omega, \qquad S^T = S, \quad \Omega^T = -\Omega.$$

Since $\boldsymbol{\omega}^T\Omega\boldsymbol{\omega} = 0$,

$$\boldsymbol{\omega}\cdot(\boldsymbol{\omega}\cdot\nabla)u = \boldsymbol{\omega}^T S\boldsymbol{\omega}. \tag{3}$$

For $\boldsymbol{\omega}\neq 0$, define $\xi = \boldsymbol{\omega}/|\boldsymbol{\omega}|$. Then

$$\frac{D|\boldsymbol{\omega}|}{Dt} = (\xi^T S\xi)|\boldsymbol{\omega}| + \nu\,\xi\cdot\Delta\boldsymbol{\omega} + \text{forcing}. \tag{4}$$

Thus the local amplitude is governed by three competing ingredients: stretching/alignment, viscous diffusion, and forcing. The scalar magnitude $|\boldsymbol{\omega}|$ alone does not determine the stretching coefficient $\xi^T S\xi$.

3 What the scalar investigation established

Let

$$Q(t) = \|\boldsymbol{\omega}(t)\|_\infty.$$

A hypothetical Riccati inequality

$$D^- Q \geq CQ^2, \qquad C > 0,$$

would imply finite-time divergence by comparison. More generally, $D^-Q \geq CQ^{1+\alpha}$ with $\alpha > 0$ gives the same qualitative conclusion.

However, no universal positive autonomous feedback $D^-Q \geq F(Q) > 0$ can hold over the full smooth class: sufficiently small smooth periodic data generate global smooth solutions with decaying vorticity, so $Q(t)\to 0$ in that regime. Independently, periodic embedded two-dimensional flows of the form

$$u(x,y,z,t) = (u_1(x,y,t), u_2(x,y,t), 0), \qquad \boldsymbol{\omega} = (0,0,\boldsymbol{\omega}_3)$$

satisfy $(\boldsymbol{\omega}\cdot\nabla)u = 0$. Hence nonzero vorticity can coexist with identically zero stretching. The periodic formulation avoids the infinite-energy issue of a z-invariant whole-space extension.

The conclusion is precise: *vorticity magnitude is a singularity observable but is not a sufficient autonomous amplification state for the full smooth class.*

4 The geometric pivot

The missing local information in (**??**) is not an arbitrary extra scalar. It is geometric:

$$\lambda = \xi^T S\xi.$$

Let e_1, e_2, e_3 be an orthonormal eigenbasis of S,

$$Se_i = \lambda_i e_i, \qquad \lambda_1 \geq \lambda_2 \geq \lambda_3, \qquad \lambda_1+\lambda_2+\lambda_3 = 0.$$

Writing $\xi = \sum_i a_i e_i$ gives

$$\lambda = \sum_{i=1}^{3} \lambda_i a_i^2. \tag{5}$$

Equation (**??**) provides a natural geometric coordinate system. Positive stretching requires sufficiently strong weight on expanding strain directions; negative or neutral stretching can occur despite large $|S|$.

This motivates a geometry audit rather than a geometry guess.

5 Dimensionless geometric coordinates

A useful local state should distinguish amplitude, strain, alignment, direction-field variation, and concentration. Candidate dimensionless quantities include

$$A_1 = \frac{\xi^T S \xi}{|S|}, \tag{6}$$

$$A_2 = \frac{\lambda_1 - (\xi^T S \xi)}{|S|}, \tag{7}$$

$$A_3 = \frac{|\nabla \xi|}{|\boldsymbol{\omega}|^{1/2}}, \tag{8}$$

$$A_4 = L\,|\nabla \xi|, \tag{9}$$

$$A_5 = \frac{|S|}{|\boldsymbol{\omega}|}, \tag{10}$$

whenever the denominators are nonzero. These are not asserted to form a complete invariant set. They are coordinates for classification.

The most basic alignment coordinate is

$$A_1 = \frac{\xi^T S \xi}{|S|} \in [-1, 1],$$

where the exact range depends on the matrix norm convention but the sign is invariant. A positive value indicates net instantaneous stretching along the vorticity direction; a negative value indicates compression.

The concentration scale from the previous state audit is

$$L^2 = \frac{E}{D}, \qquad E = \frac{1}{2}\int |\boldsymbol{\omega}|^2 dx, \qquad D = \int |\nabla \boldsymbol{\omega}|^2 dx.$$

A candidate singular family must explain not only how amplitude grows, but how its characteristic scale changes.

6 A first geometric classification

We propose four broad classes. They are classification hypotheses, not theorems.

6.1 Amplifying configurations

These satisfy sustained positive stretching in the vorticity direction,

$$\lambda = \xi^T S \xi > 0,$$

with sufficient persistence that the positive production is not erased by diffusion or geometric depletion. A necessary local signature is therefore persistent positive contribution in (**??**).

6.2 Neutral configurations

These satisfy approximately

$$\xi^T S\xi \approx 0.$$

Embedded two-dimensional flows are exact examples. Such configurations may possess large vorticity but do not obtain amplitude growth from vortex stretching.

6.3 Depleting configurations

These satisfy

$$\xi^T S\xi < 0$$

for sufficiently long intervals or with sufficient magnitude to offset any positive episodes. Such configurations are natural candidates for regularizing families.

6.4 Intermittently amplifying configurations

These alternate between positive and negative stretching. Instantaneous positivity is then insufficient; cumulative amplification must be assessed. A natural quantity is

$$\mathcal{A}(t_0, t) = \int_{t_0}^{t} \xi^T S\xi \, ds.$$

The challenge is to compare cumulative stretching with cumulative viscous depletion and concentration effects.

7 Necessary conditions for a candidate singular family

A candidate finite-time singularity at time T must satisfy more than $Q(t) \to \infty$. A geometry-based necessary-condition checklist is:

G1. **Amplitude:** $Q(t)$ must become unbounded or otherwise violate the appropriate continuation criterion.

G2. **Positive stretching:** the vorticity direction must experience sufficiently persistent positive $\xi^T S\xi$.

G3. **Alignment persistence:** the vorticity direction must maintain an adequate relation to the expanding strain eigenspaces.

G4. **Concentration:** the characteristic scale L or an equivalent critical concentration measure must contract sufficiently rapidly.

G5. **Viscous balance:** stretching must overcome the relevant diffusive contribution rather than merely dominate instantaneously.

G6. **Directional coherence:** the geometry must avoid whatever directional depletion is sufficient to prevent singularity.

G7. **Pressure compatibility:** the pressure Hessian must be dynamically compatible with the proposed strain evolution.

G8. **Persistence:** all required inequalities must remain compatible as $t \uparrow T$.

None of these conditions alone is sufficient. Their simultaneous persistence is the actual geometric problem.

8 Stretching versus viscosity

From (**??**), a local candidate amplification balance is

$$\lambda|\boldsymbol{\omega}| \quad \text{versus} \quad -\nu\,\xi\cdot\Delta\boldsymbol{\omega}.$$

At a point where $|\boldsymbol{\omega}|$ is locally maximal, one cannot simply replace the diffusive term by a fixed negative multiple of $|\boldsymbol{\omega}|^2$ without additional geometric control. This is one reason the concentration variable is important.

At the global level,

$$E' = P - \nu D.$$

Thus a sustained singular route must reconcile production P with dissipation νD. The ratio

$$\Pi = \frac{P}{\nu D}$$

when defined, is a useful diagnostic: $\Pi > 1$ means net enstrophy growth at that instant, but $\Pi > 1$ alone is not a blow-up criterion.

9 Why large strain is not sufficient

Large $|S|$ does not imply large positive $\xi^T S\xi$. For example, a vortex configuration can induce strong strain while its vorticity is oriented in a direction with zero or negative quadratic form. Idealized antiparallel vortex tubes illustrate the distinction between strain magnitude and self-stretching.

Likewise, a prescribed linear strain such as

$$u_s = (-a(t)x, -a(t)y, 2a(t)z)$$

with $a(t) \sim (T-t)^{-1}$ cannot establish spontaneous Navier–Stokes blow-up if the singularity is inserted through the prescribed coefficient or forcing. A valid singularity mechanism must emerge from the coupled PDE dynamics.

10 Directional geometry and depletion

Geometric regularity theory shows that vorticity direction is a meaningful regularity variable. Constantin and Fefferman established regularity criteria based on the direction of vorticity and its coherence, while Constantin–Fefferman–Majda developed related geometric constraints for Euler. These results do not imply that every successful Navier–Stokes proof must identify a singular geometry. They do establish that geometry can constrain stretching strongly enough to affect regularity.

Our use of this literature is therefore deliberately limited: it motivates directional coherence as a coordinate in the geometry audit. We do not claim equivalence between the present state-space formulation and the Constantin–Fefferman criteria.

11 A depletion family program

The complementary research target is to define broad classes of configurations that cannot sustain blow-up.

11.1 No-stretching family

If

$$\xi^T S\xi = 0$$

throughout the relevant high-vorticity region, then vortex stretching contributes nothing to local amplitude growth. Embedded 2D flows provide a concrete periodic realization.

11.2 Uniformly depleted alignment family

Suppose a class admits an estimate of the form

$$\xi^T S\xi \le \eta(t)|S|,$$

where the geometric depletion factor η is sufficiently small in a norm compatible with a known regularity criterion. The task is to determine quantitative thresholds that imply regularity. This should be connected to, not confused with, established geometric regularity criteria.

11.3 Directional roughening/decorrelation family

If the vorticity direction changes rapidly enough that positive alignment cannot persist, then cumulative stretching may fail to remain positive. The precise mechanism must be formulated analytically; visual twisting alone is not sufficient.

11.4 Insufficient-concentration family

If a candidate configuration cannot make L or another critical scale shrink at a rate compatible with the required amplification, it may be excluded as a blow-up family. This is a necessary-condition program, not an immediate regularity theorem.

12 A candidate blow-up family program

The dual objective is to characterize the remaining configurations. A candidate family should be defined by invariant conditions rather than a single picture. A provisional template is

$$\mathcal{B} = \left\{ u : \begin{array}{l} Q \to \infty, \\ \xi^T S\xi > 0 \text{ on the dominant high-vorticity set}, \\ \text{alignment persists}, \\ L \to 0 \text{ at a compatible rate}, \\ P/\nu D \text{ remains sufficiently large}, \\ \text{pressure Hessian remains dynamically compatible} \end{array} \right\}. \tag{11}$$

This is a research definition, not a theorem. The immediate task is to replace qualitative phrases by scale-invariant quantitative conditions.

13 Toward a geometric normal form

The eventual goal is not necessarily to identify one exact vortex picture. It is to derive a *normal form* for any admissible singularity. Such a normal form could specify asymptotic behavior of:

1. vorticity magnitude;
2. strain eigenvalues and their ratios;
3. vorticity–strain alignment coefficients a_i^2;

4. concentration scale;
5. directional coherence;
6. pressure-Hessian eigenstructure;
7. ratios of stretching to dissipation;
8. spatial anisotropy and localization.

If successful, the output would be a family of geometries described by invariant asymptotic relations rather than by a single coordinate-dependent configuration.

14 A useful bifurcation: blow-up-compatible versus regularity-compatible geometry

The classification can be viewed as a bifurcation problem:

$$\mathcal{G} = \mathcal{G}_{\text{regular}} \cup \mathcal{G}_{\text{candidate}}.$$

The objective is progressively to enlarge $\mathcal{G}_{\text{regular}}$ by proving exclusion theorems and thereby shrink $\mathcal{G}_{\text{candidate}}$.

A successful chain would have the form

$$\text{all smooth configurations} \supset \mathcal{G}_1 \supset \mathcal{G}_2 \supset \cdots \supset \mathcal{G}_{\text{candidate}},$$

where each elimination is a mathematically justified necessary-condition argument.

The reverse direction can then test whether the remaining candidate family is dynamically realizable.

15 The role of the pressure Hessian

The pressure Hessian is a central obstacle to purely local geometry. The strain equation has the form

$$\frac{DS}{Dt} = -\nabla^2 p + \nu \Delta S - (S^2 + \Omega^2).$$

Thus even if the local alignment data are specified, the surrounding flow influences strain evolution through pressure. Any claimed geometric normal form must therefore either:

1. derive a compatible pressure-Hessian asymptotic structure;
2. show that pressure contributions are lower order in the candidate regime; or
3. identify a cancellation or monotonicity mechanism that controls them.

Ignoring the pressure Hessian would repeat the closure error identified in v2.

16 Scale invariance and criticality

Under Navier–Stokes scaling,

$$u_\lambda(x,t) = \lambda u(\lambda x, \lambda^2 t), \qquad \boldsymbol{\omega}_\lambda(x,t) = \lambda^2 \boldsymbol{\omega}(\lambda x, \lambda^2 t).$$

Any proposed geometric criterion should therefore be tested for compatibility with this scaling. A configuration family expressed only through dimensional quantities can be misleading. The preferred formulation is through dimensionless ratios or critical norms.

This is especially important for the concentration variable L: its absolute value depends on units, whereas ratios comparing L with a dynamically generated scale can be invariant.

17 Relation to established geometric regularity theory

The present framework is complementary to established results rather than a replacement for them. Constantin–Fefferman type criteria show that sufficiently coherent vorticity direction can prevent the mechanism required for singularity. Later work has developed geometric and depletion criteria in multiple formulations. Surveys of geometric constraints emphasize that regularity criteria can be interpreted as restrictions on any possible blow-up configuration.

Our proposed geometry audit adopts that philosophy but asks a different organizational question: can those restrictions be assembled into a classification architecture in which configuration families are progressively eliminated until only blow-up-compatible states remain?

This note therefore treats established geometric criteria as *exclusion tools* within a broader state-space program.

18 A precise research algorithm for v3 and beyond

The next mathematical passes should proceed in the following order.

1. **Define the candidate high-vorticity set.** Avoid relying solely on a single pointwise maximum; introduce suitable superlevel sets or localized concentration measures.

2. **Diagonalize strain locally.** Express ξ in the strain eigenbasis and retain the coefficients a_i^2.

3. **Normalize.** Construct dimensionless alignment and concentration parameters.

4. **Derive exact evolution identities.** Do not replace pressure, diffusion, or geometric terms by heuristic estimates prematurely.

5. **Prove exclusion lemmas.** Identify geometric assumptions that force boundedness, depletion, or decay.

6. **Compare with known regularity criteria.** Determine which candidate families are already excluded by Prodi–Serrin, geometric depletion, localized criteria, or endpoint results.

7. **Characterize the remainder.** Express the surviving candidate family through invariant asymptotic inequalities.

8. **Test realizability.** Determine whether the full Navier–Stokes equations admit a trajectory that remains inside the candidate family.

9. **Only then investigate blow-up.** A candidate configuration is not a singularity proof until existence and asymptotic consistency are established.

19 The central research question

The next-stage question can now be stated precisely:

> Can one characterize a necessary geometric normal form for any finite-time singularity of the three-dimensional incompressible Navier–Stokes equations by progressively eliminating geometric configurations that cannot sustain the coupled amplification–concentration–alignment feedback required for blow-up?

The complementary question is:

> Can one construct broad, mathematically defined families of geometric configurations for which vortex stretching is sufficiently depleted, concentration is insufficient, or pressure/strain dynamics are incompatible with finite-time singularity?

These two questions define the Geometry Audit program.

20 What would constitute a major result?

There are several levels.

20.1 Level I: exclusion theorem

Prove that a broad geometric family cannot contain a singularity.

20.2 Level II: necessary geometric condition

Prove that every singularity must satisfy a quantitative alignment/concentration condition.

20.3 Level III: geometric normal form

Show that every possible singularity is asymptotically equivalent, in an invariant sense, to a restricted family of configurations.

20.4 Level IV: nonexistence or construction

Either prove that the remaining family is dynamically impossible, yielding regularity, or construct a genuine Navier–Stokes solution exhibiting the required finite-time singularity.

The present note targets Levels I–II and establishes the framework for Level III. It does not claim Level IV.

21 Comparison with configuration-first approaches

A configuration-first strategy starts from a particular geometry and asks whether it can produce the required singular behavior. The state-first strategy developed here starts from the amplification observable and asks what variables must be controlled before geometry can be inferred.

The contrast can be summarized as

$$\boxed{\text{configuration} \to \text{dynamics}} \qquad \text{versus} \qquad \boxed{\text{state} \to \text{constraints} \to \text{configuration}}.$$

The present work does not assert that the second strategy is universally superior. Its advantage is methodological: it offers a route to eliminate entire configuration classes without committing to a particular singular geometry in advance.

22 Limitations

Several limitations must remain explicit.

1. The classification variables are not yet proven to be complete invariants.
2. Positive stretching is necessary for a simple vorticity-amplitude growth mechanism but is not by itself sufficient for singularity.

3. The concentration scale $L^2 = E/D$ is useful but is not a complete description of spatial localization.
4. Pressure remains nonlocal and prevents purely local closure.
5. Existing geometric regularity criteria cannot simply be reinterpreted as a complete classification theorem.
6. The surviving candidate family may still be empty, nonempty, or too broad; this is an open research question.

23 Conclusion

The progression from the scalar investigation to the geometry audit is a natural continuation of the same research program. The first stage established that vorticity magnitude alone cannot furnish a universal positive scalar closure. The second stage showed that adding the natural strain-aware production variable does not close the dynamics because pressure, nonlocality, concentration and higher-order information re-enter.

The third-stage pivot is therefore to classify the geometric states themselves. The central object is not a single vortex picture but a family described by invariant relations among vorticity direction, strain eigenstructure, alignment, concentration, directional coherence, dissipation, and pressure-Hessian dynamics.

The desired convergence is

$$\text{all configurations} \longrightarrow \text{provably regular/depleted families} \longrightarrow \text{remaining candidate families} \longrightarrow \text{geometric norn}$$

A successful outcome would not merely guess a singular geometry. It would derive the constraints that any singular geometry must satisfy.

The key methodological takeaway is:

> **Do not guess the geometry first. Discover the information state that the equations require, use that state to eliminate impossible configurations, and let the surviving geometry emerge as a consequence of the PDE constraints.**

Claim boundary

This research note does not solve the Navier–Stokes Millennium Prize problem. It does not prove global regularity or finite-time blow-up. It does not prove that every successful proof must explicitly construct a singular geometry. Its contribution is a structural and methodological framework for classifying and eliminating geometric configuration families after scalar closure has failed.

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